CS & IT ENGINEERING

Digital Logic

Optimisation

Lecture No. 03

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Recap of Previous Lecture

Topic

V. Imp

Min no. of Universal Gates.

Topics to be Covered

Topic

Few more Cases

→ Application

Case 1 :- $A \cdot B \cdot \bar{C}$

NAND	NOR
$[(2n-2)+K]$	$[(3n-3)-K]$

Case 2 :- $\bar{A}+\bar{B}+C$

NAND	NOR
$[(3n-3)-K]$	$[(2n-2)+K]$

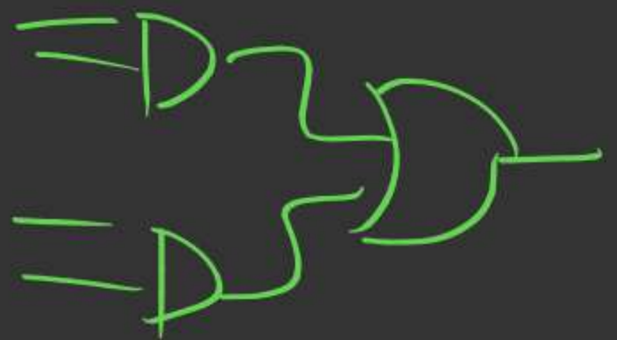
Case 3:- SOP

$AB + CD + XY$

$AB + C$

$\downarrow$

$AB + C \cdot C$



AOI

2nd i/p

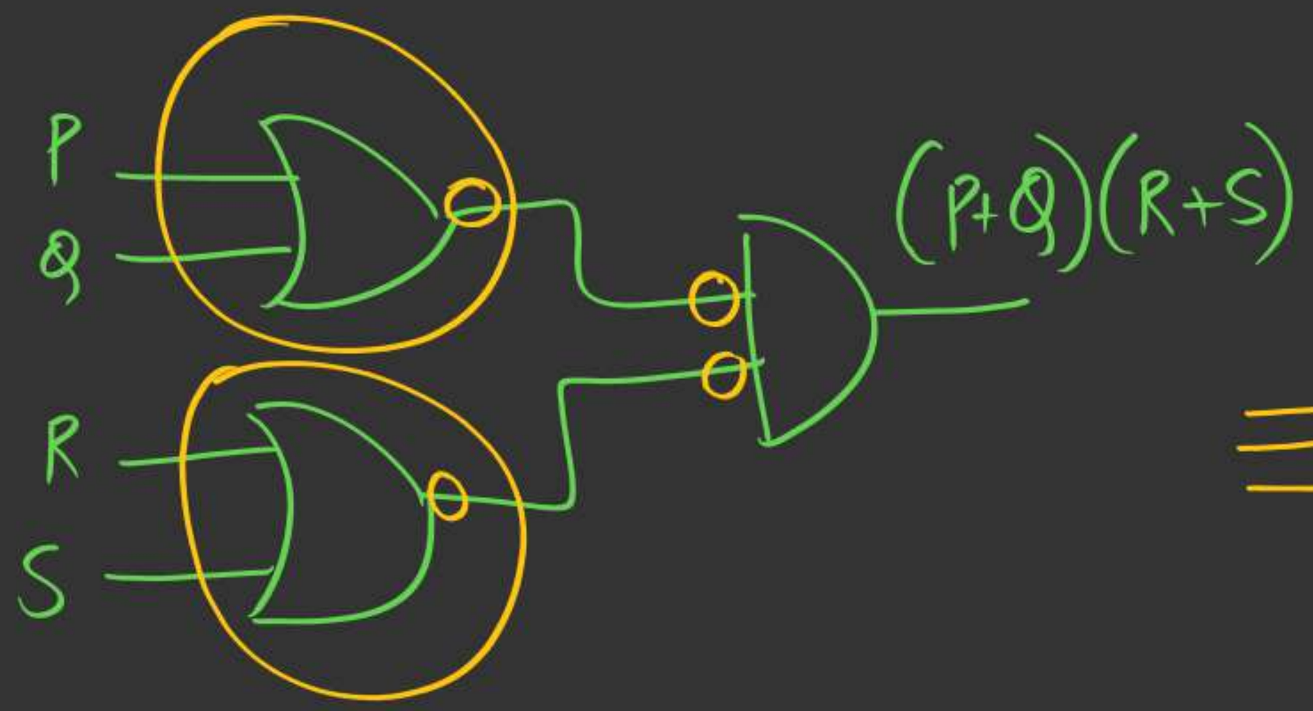
NAND.

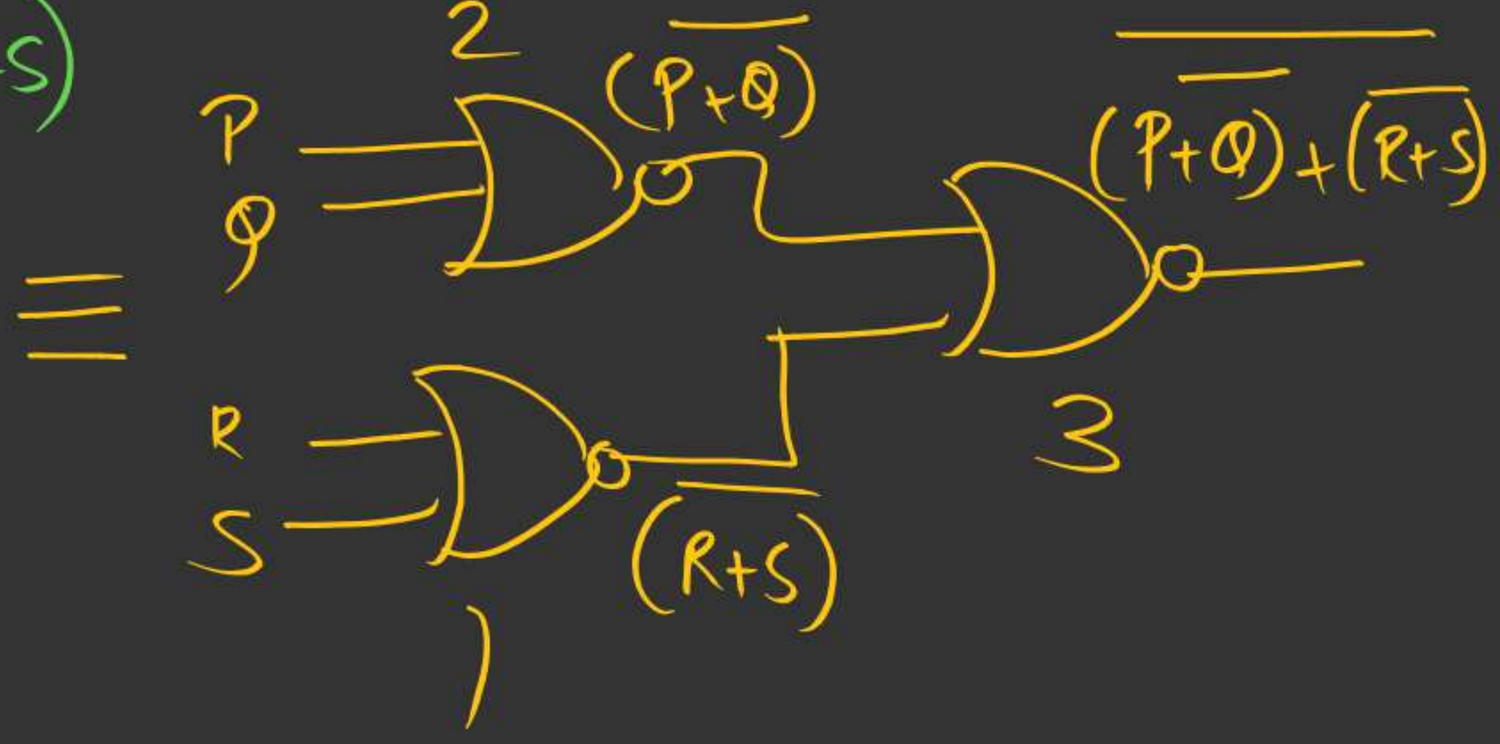
Case 4:- POS $\longrightarrow$ product of Sums (NOR)

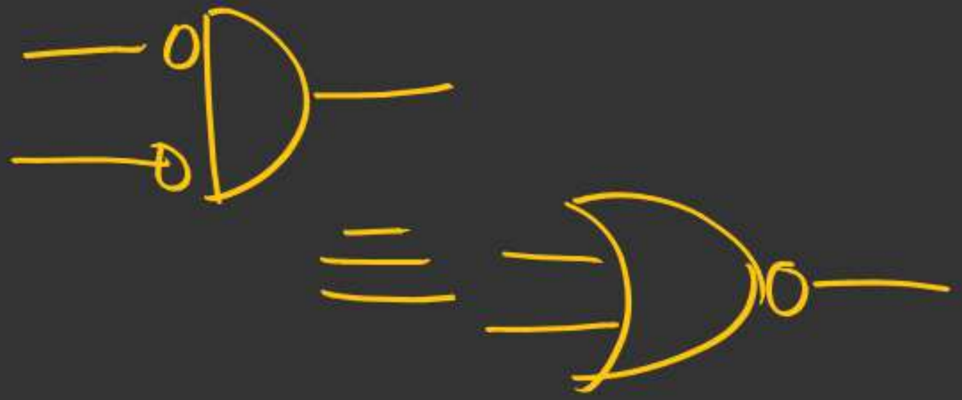
eg:- $y = (P+Q)(R+S)$

$y = (P+Q)(R+S)$

AOI







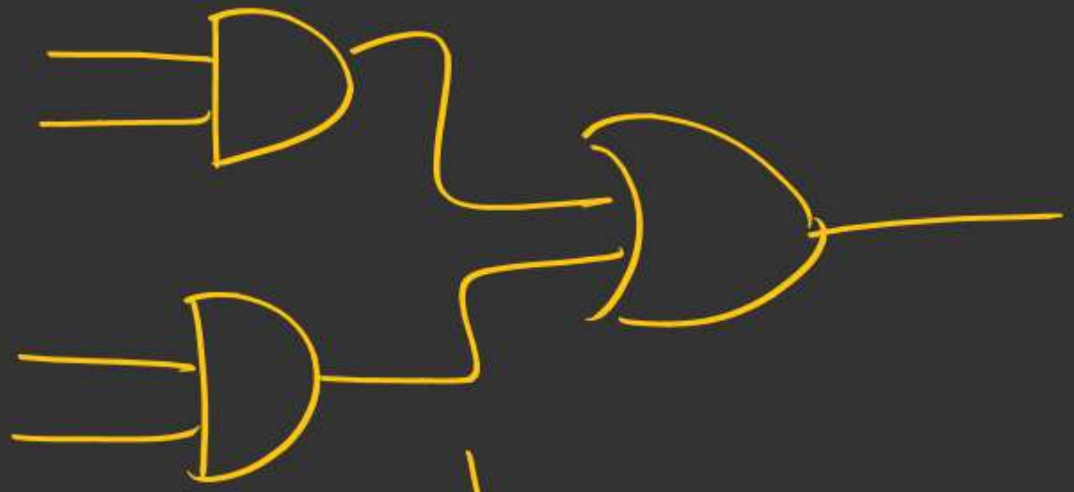
$\boxed{\text{Ans: 3}}$

$$y = \overline{\overline{(P+Q)} + \overline{(R+S)}}$$

$$= \overline{\overline{(P+Q)}} \cdot \overline{\overline{(R+S)}}$$

$$y = \underline{\underline{(P+Q) \cdot (R+S)}}$$

SOP



AND	OR
NAND	NAND

POS

OR	AND
NOR	NOR

Imp:

$\boxed{SOP} \rightarrow$ NAND

$\boxed{POS} \rightarrow$ NOR

Min no. of NOR?

1) Convert to POS.

2) Implement using AOI

OR $\longrightarrow$ AND

3) Convert NOR $\rightarrow$ NOR $\longrightarrow$ Count NORs

eg:- $y = PQ + R$

min NAND $\longrightarrow$

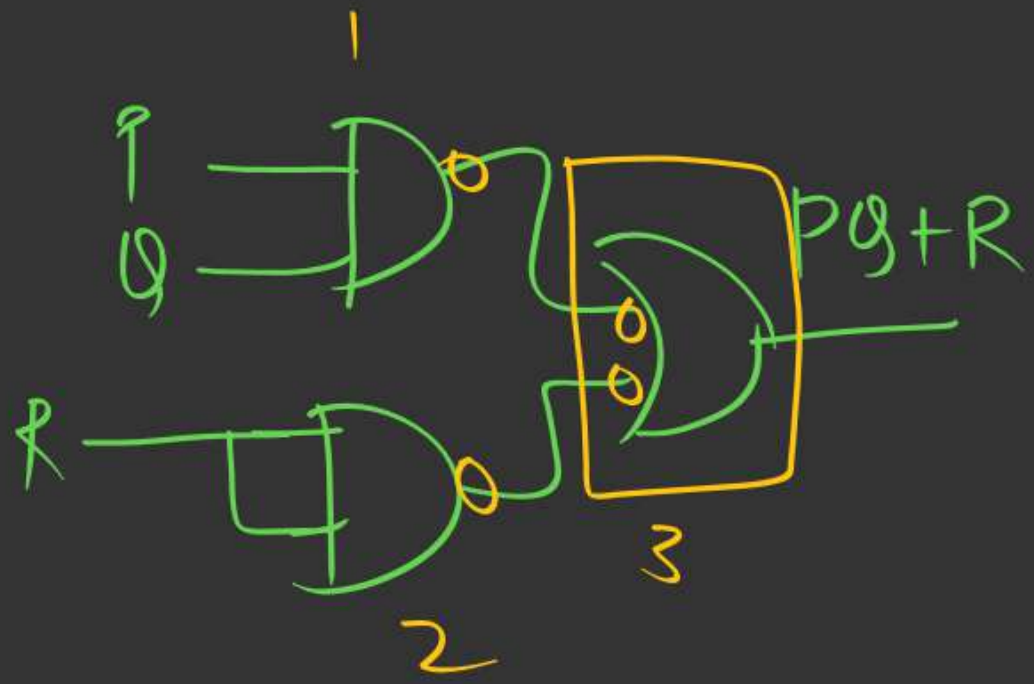
min NOR ?

$$y = PQ + R$$

NAND → <u>SOP</u>

3

$$y = PQ + R$$
$$= PQ + R \cdot R$$



$y = PQ + R$

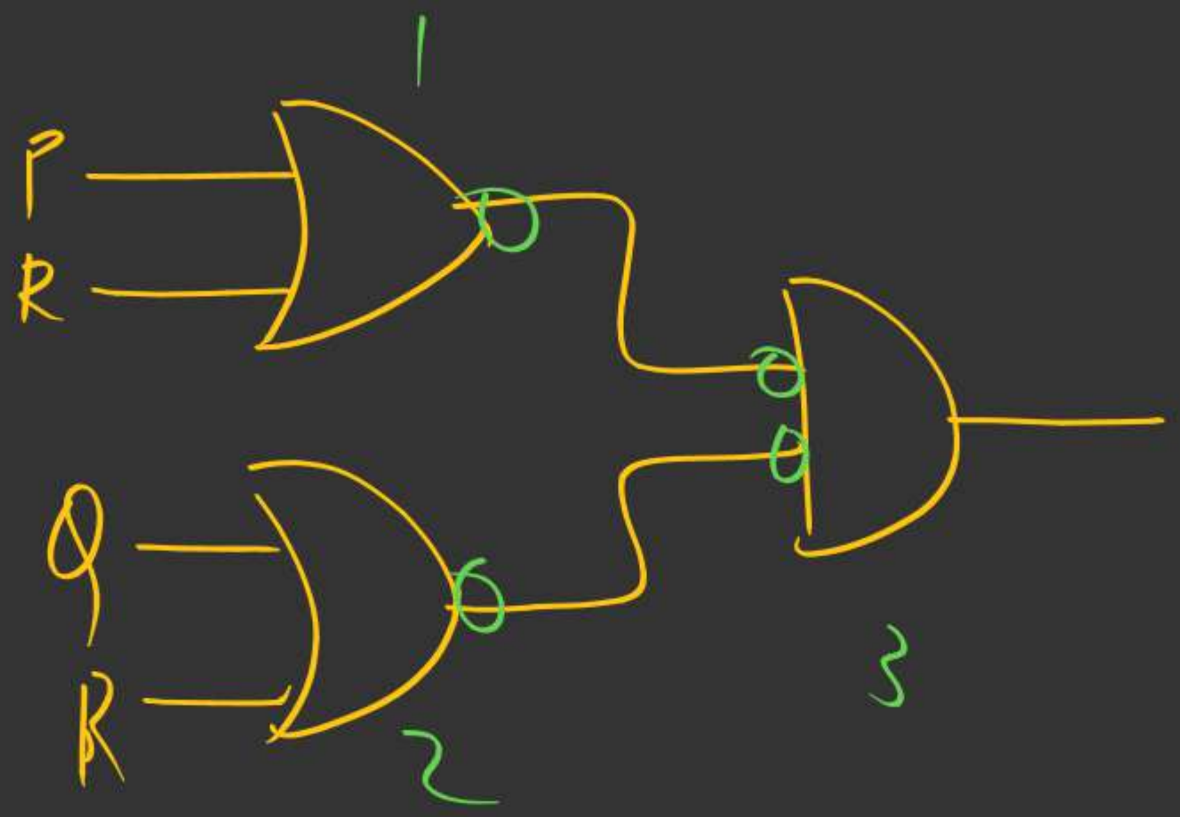
NOR $\rightarrow$ POS

$\downarrow$

3

$y = R + PQ$

$= (R+P)(R+Q)$



AOI

H.W

$$\boxed{y = AB + BC + CD + DA}$$

$$= B(A+C) + D(A+C)$$

NAND

Let $A+C=P \rightarrow 3$

$$= \boxed{B\cdot P + D\cdot P} \longrightarrow 3$$

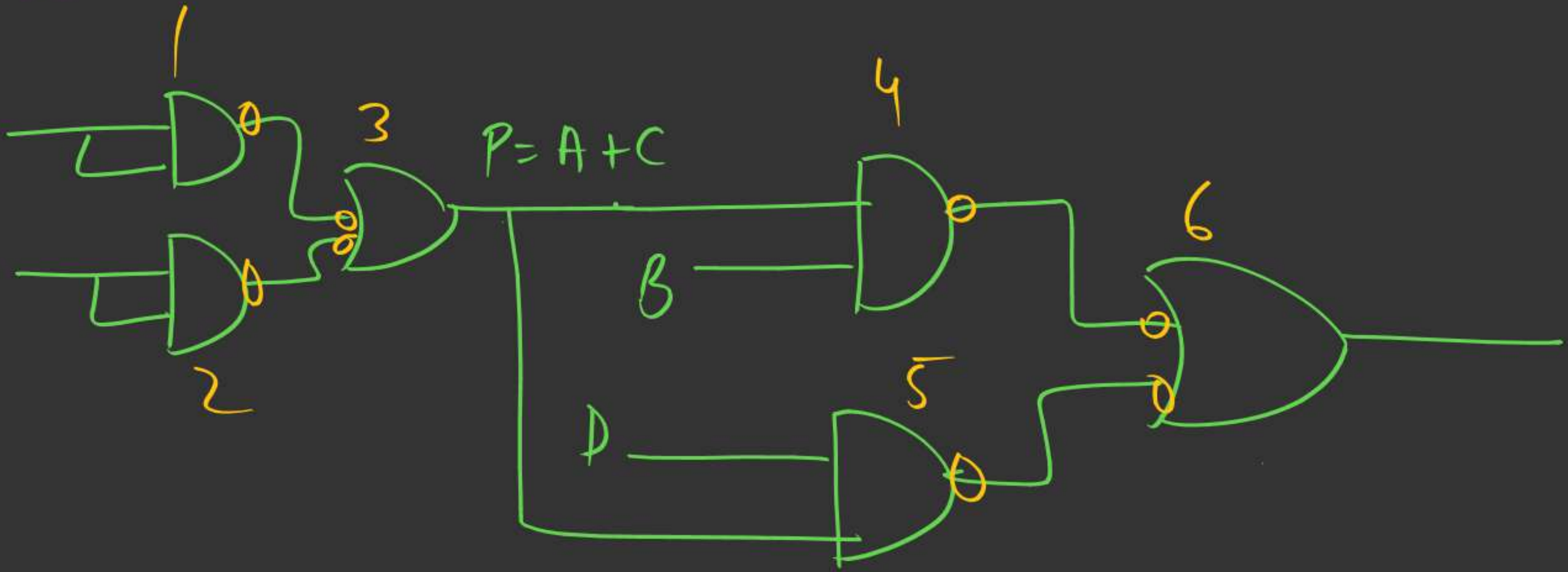
9

8

6

$P = A+C$

$= \overline{\overline{A}\cdot\overline{A}} + \overline{\overline{C}\cdot\overline{C}}$



Ans:- 6

exception $\longrightarrow$ XOR

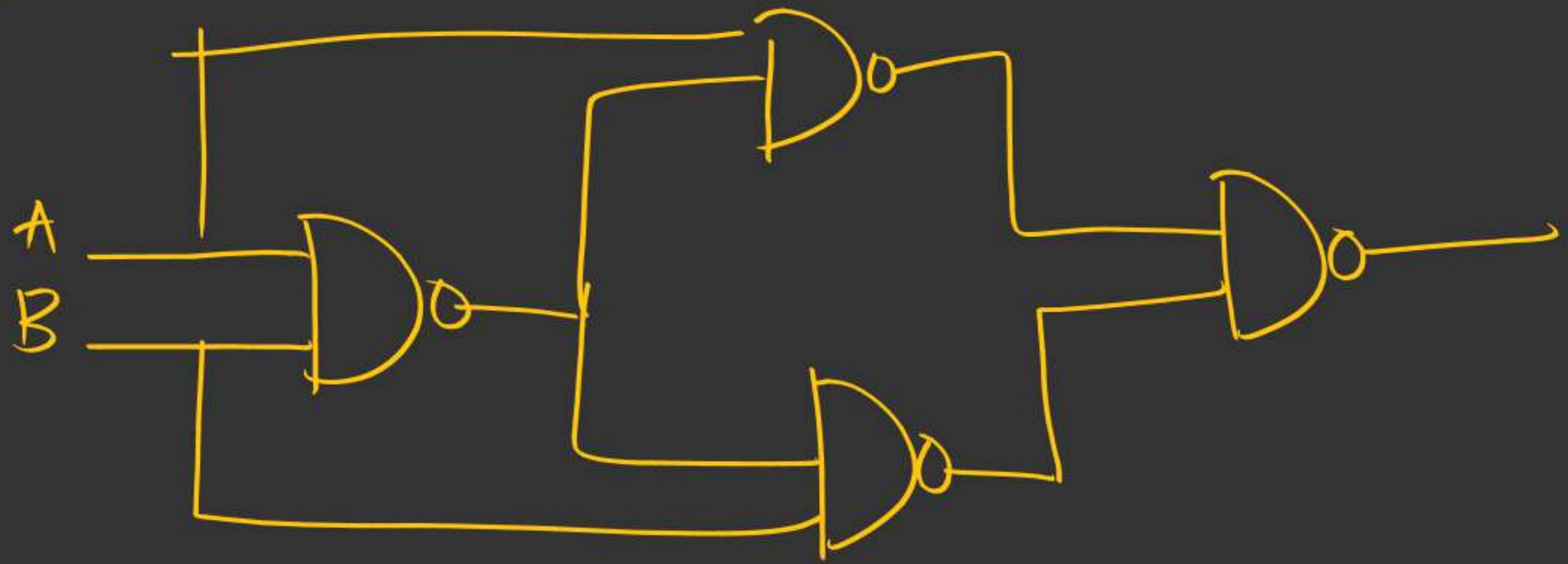
XOR

$$\boxed{y = A\bar{B} + \bar{A}B}$$

$$= A \oplus B$$

4 NAND

5 NOR



MSQ

(Q) $f(P,Q) = P \oplus Q P \oplus Q$

f is equivalent to ?

A) NAND

B) OR

C) min 2 NAND gates reqd

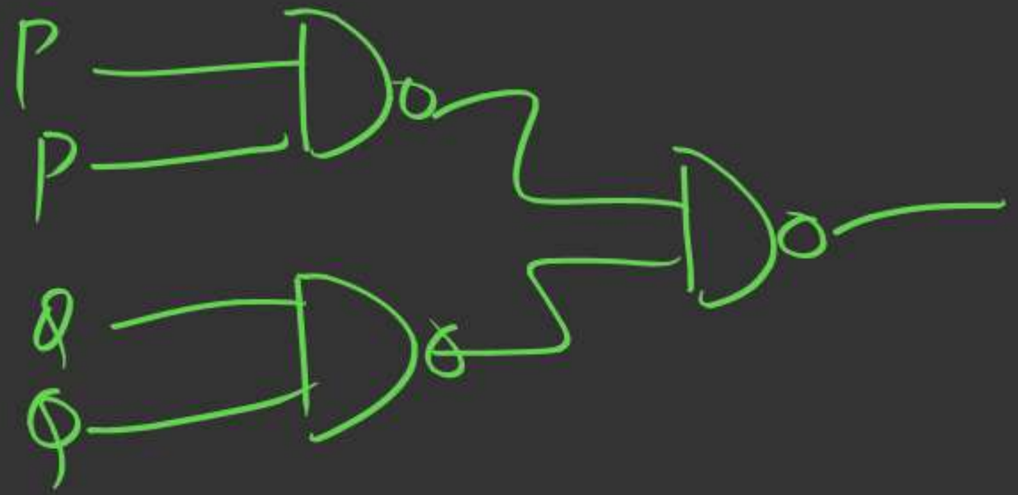
D) min 3 NAND " "

Soln :

$$F = P \oplus Q P \oplus Q$$

$$F = P \oplus Q \oplus PQ$$

$$= \underline{\underline{P+Q}}$$



Appr2 :-

(Backup)

P	Q	$P.Q$	$f = P \oplus PQ \oplus Q$
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	1

OR

(Q.2) $F(P,Q) = \underbrace{\overline{(P+Q)}}_{X} \oplus P$

A) $P \cdot Q$

B) $\overline{P} Q$ ✓

C) $\overline{Q} P$

D) $P+Q$

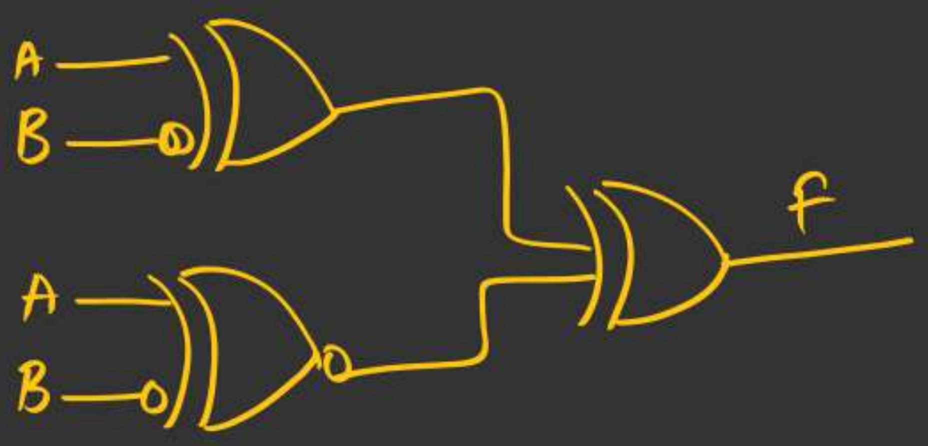
$F = X \oplus P$

$= \overline{X} P + \overline{P} X$

$= \overline{\overline{(P+Q)}} \cdot P + \overline{P} \cdot (P+Q)$

$= \overline{P} \cdot \cancel{\overline{Q} \cdot P} + \cancel{\overline{P} \cdot P} + \overline{P} Q = \boxed{\overline{P} Q}$

(Q.3)



50%

A) 0

B) $A \oplus B$

C) $A \odot B$

D) 1

$$f = (A \oplus \bar{B}) \oplus (A \odot \bar{B})$$

$$= \underbrace{(A \odot B)}_{P} \oplus \underbrace{(A \oplus B)}_{\bar{P}} = P \oplus \bar{P}$$

$$= 1$$

(Q.1) $F(P,Q,R) = Q \cdot P \cdot \overline{R} + P + R \cdot P \cdot Q$

min NAND gates?

26%

$F(P,Q,R) = Q \cdot P \cdot \overline{R} + P + RPQ$

$= P(Q\overline{R} + 1 + QR)$

$= P$

Ans. $\boxed{0}$

P — P

(9)

$$F = P + P\overline{Q}$$

$$= P(1 + \overline{Q})$$

$$= P$$

(g)

$$
\begin{aligned}
F(P,Q) &= \bar{P}\cdot\bar{Q} + P\cdot Q + \bar{P}\cdot Q \\
&= \bar{P}(\bar{Q}+Q) + PQ \\
&= \bar{P}\cdot 1 + P\cdot Q \\
&= \bar{P} + P\cdot Q \\
&= (\cancel{\bar{P}+P})(\bar{P}+Q)
\end{aligned}
$$

$$F(P,Q) = \underline{\underline{(\bar{P}+Q)}}$$

(Q) $F(P,Q) = \bar{P}\cdot Q + \bar{P}\cdot\bar{Q} + P\cdot Q + P\cdot\bar{Q}$

$= \bar{P}(Q+\bar{Q}) + P(Q+\bar{Q})$

$= \bar{P}+P$

$= 1$

minterms

$F = (\bar{P}\cdot Q + P\bar{Q}) + PQ + \bar{P}\bar{Q}$

$= \underbrace{(P \oplus Q)}_{A} + \underbrace{(P \odot Q)}_{\bar{A}}$

$= A + \bar{A}$

$= 1$

n=1 variable :-

P
P
1
0

$\rightarrow$ 4 ✓

$\underline{n=2}$ i/p var

How many simplied O/ps ? $\rightarrow \underline{16}$

$\underline{P, Q}$		(SOP) $\underline{\text{minterms}}$	(POS terms) $\underline{\text{maxterms}}$
1	P	$P \cdot Q$	$P + Q$
0	$\overline{P}$	$P \cdot \overline{Q}$	$P + \overline{Q}$
	Q	$\overline{P} \cdot Q$	$\overline{P} + Q$
2	$\overline{Q}$	$\overline{P} \cdot \overline{Q}$	$\overline{P} + \overline{Q}$
	4	4	4

$\overline{P} \cdot \overline{Q} + PQ = P \odot Q$

$\overline{P} Q + P \overline{Q} = P \oplus Q$

2

16

In general,

'n' i/p variables

$2^{(2^n)}$ distinct expressions can be formed

$(2^n \rightarrow \text{minterms})$ $(2^n \rightarrow \text{maxterms})$

(Q) $y = A + BC + \bar{A}C = ?$

Options → PDF

Lec-4 HW (24th April)

open PDF

(Q.1) $PQ + Q\bar{R} + \bar{P}R$ simplies to ?

P ✗
Q ✗
R ✗

A) $(\bar{P}+Q)(Q+R)$ ✓

B) $PQ + Q\bar{R}$

C) $PQ + \bar{P}R$

D) $\bar{P}R + Q\bar{R}$

$$PQ + Q\bar{R} + \bar{P}R$$

$$Q(P+\bar{R}) + \bar{P}.R = Q \cdot X + \bar{X}$$

$$= (Q+\bar{X})(X+\bar{X})$$

$$= (Q+\bar{X})$$

Let $P+\bar{R} = X$

$$\bar{X} = \overline{P+\bar{R}}$$

$$= \bar{P}.R$$

$$y = Q + \overline{X}$$

$$\overline{X} = \overline{P} \cdot R$$

$$= \underline{\underline{Q + \overline{P}\,R}}$$

$$= (Q + \overline{P})(Q + R)$$

$SOP \longrightarrow POS$

$POS \longrightarrow SOP$

$F \qquad \overline{F'} \longrightarrow SOP$

MSQ

(Q.2) $F(P,Q,R) = \sum(1,2,4,6,7)$

then $F(P,Q,R)$ is ?

A) $\prod(0,3,5)$

B) $\sum(0,3,5)$

C) $\sum(1,2,4,6,7)$

D) $\prod(1,2,4,6,7)$

Ans B, D

H.W

$$y = (P + Q + RS)(\bar{P} + Q + \bar{R} \cdot S)(\bar{P} + Q + R)$$

Simplify

2 mins Summary

→ Min universal Gates

→ Imp Practice Questions

THANK - YOU